Midterm Practice Questions · Answer Key

MATH60033A · Quantitative Methods in International Business

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Open this only after answering the questions on paper. The answers are model answers: on the exam, any precise formulation earns the marks, and **working is marked** — a correct derivation with an arithmetic slip earns most of them.

Part A — Definitions and short answers

A1. The mean is the sum divided by n, the balance point; the median is the middle value once sorted. The median is robust: an extreme value moves the mean (it enters the sum) but not the middle position.

A2. The mean computed after dropping a fixed share (say 5% or 10%) of observations from each tail. Report it when a few extreme values distort the mean but you want to use more of the data than the median does.

A3. The deviations from $\bar{x}$ sum to zero, because $\bar{x}$ was estimated from the same data: only $n-1$ of them are free. Dividing by n would systematically underestimate the variance.

A4. The SD measures the spread of the **observations**; the SE, $s/\sqrt{n}$, measures the spread of the **mean** across repeated samples. A larger sample shrinks the SE (the mean is estimated more precisely) but not the SD (the observations are as spread out as before).

A5. $IQR = Q_3 - Q_1$, the width of the middle 50% of the data. It ignores both tails, so a long tail does not inflate it the way it inflates the SD.

A6. A long **right** tail: a few large values. They pull the mean above the median.

A7. How heavy the tails are compared with a normal distribution: $g_2 > 0$ means more extreme values than a normal. A normal has excess kurtosis 0.

A8. Not necessarily. The rule holds for a roughly **normal** variable; a symmetric distribution can still have heavy or light tails (check g_2).

A9. The sum of squared residuals, $\sum_i (y_i - \hat{y}_i)^2$. Squaring makes every miss positive, so misses above and below cannot cancel, punishes one large miss more than several small ones, and gives a closed-form solution, $(X^\top X)^{-1} X^\top y$.

A10. "Comparing countries with the same investment, a one-percentage-point higher tertiary education share is associated with a 0.31-percentage-point higher GDP growth." Units, "holding investment fixed", and "associated with", not "causes".

A11. $\hat{\beta}_j$ equals the slope from regressing the part of y not explained by the other regressors on the part of x_j not explained by them. It is a **partial** relationship: what is left of the association after the other regressors have taken their share.

A12. $R^2 = 1 - SSR/SST$, the share of the variance of y the model accounts for. It does not say the relationship is causal, nor that the model's assumptions hold (a curved cloud can have a high R^2), nor that it will predict new data well.

A13. R^2 never falls when a regressor is added, even a useless one. The adjusted R^2 penalises each extra regressor and falls when the new one improves the fit by less than chance would.

A14. $RSE = \sqrt{SSR/(n-k-1)}$: the typical size of a residual, in the **units of** y — the typical miss of the model.

A15. Linearity in the parameters, exogeneity $\mathbb{E}[\varepsilon \mid X] = 0$, and no perfect collinearity (full column rank). Constant variance (homoskedasticity) is **not** needed for unbiasedness — only for the classical standard errors and for efficiency.

A16. The variance of the errors changes with the fitted values: the residual plot **fans out**. The coefficients stay unbiased, but the classical standard errors are wrong, so tests and intervals are wrong. Use robust (HC) standard errors.

A17. An **outlier** has a large residual (unusual y given x). A **leverage point** has an unusual x. An **influential point** changes the fit when removed — typically high leverage **and** a large residual.

A18. How much all the fitted values move when that observation is deleted; it combines leverage and residual. Investigate it — find which row it is and why — and report the fit with and without it. Never delete it silently.

A19. $VIF_j = 1/(1 - R_j^2)$, where R_j^2 is from regressing x_j on the other regressors. Above about 10, x_j is almost a combination of the others: its standard error is inflated and its sign can be unstable (multicollinearity).

A20. BIC, because its penalty per parameter, $\ln n$, exceeds AIC's 2 as soon as $n > 7$. The models must be fitted on **the same rows**; otherwise the likelihoods are not comparable.

A21. It can predict probabilities below 0 or above 1, and it forces the effect on the probability to be the same everywhere (a straight line), which is implausible near 0 and 1. Its errors are also heteroskedastic by construction.

A22. odds $= p/(1-p)$. For $p = 0.20$: $0.20/0.80 = 0.25$, "one event for every four non-events".

A23. In **log-odds**: one more unit of x raises the log-odds by 0.26. $e^{0.26} = 1.30$ is the **odds ratio**: the odds are multiplied by 1.30, i.e. 30% higher. The wrong reading: "the **probability** is 30% higher".

A24. The probability is an S-shaped function of x: its slope, $p(1-p)\beta$, depends on where you evaluate it, largest at $p = 0.5$ and small near 0 or 1. A constant odds ratio moves the probability by different amounts depending on the starting point.

A25. The parameter values that make the observed data — here, the 0s and 1s actually observed — as probable as possible.

A26. Wald: $z = \widehat{\beta}/SE$, one coefficient at a time, from one model. Likelihood ratio: $LR = 2(\ell_{full} - \ell_{restricted}) \sim \chi^2_q$, comparing two nested models. Use the LR test for several coefficients **jointly**, such as all the dummies of one categorical variable.

A27. $1 - \ell_{model}/\ell_{null}$: how far the model's log-likelihood improves on the intercept-only model. It is not a share of variance, and its values run lower than an OLS R^2; 0.2–0.4 already indicates a good fit.

A28. The cutpoints $c_1 < c_2 < ...$ are the thresholds between categories, one per cut; they act as intercepts. Proportional odds: each predictor has **one** coefficient, so the same odds ratio applies at every cut. Check it by fitting a binary logit at each cut and comparing the slopes.

A29. The category every other one is compared with; each equation models $\ln[P(k)/P(\text{baseline})]$. An RRR of 0.80: a one-unit rise in x multiplies the ratio $P(\text{vocational})/P(\text{academic})$ by 0.80 — 20% fewer vocational students per academic one. It does not say the probability of vocational falls by 20%.

A30. Discrimination: does the model **rank** positives above negatives (AUC)? Calibration: when the model says 20%, does the event happen about 20% of the time? AUC depends only on the ranking, so a model can rank perfectly while every probability is, say, twice too high — useless for pricing or thresholds.

A31. With a 2% base rate, predicting "never" is 98% accurate and finds nothing. Precision $= TP/(TP + FP)$: of those flagged, the share that are positive. Recall $= TP/(TP + FN)$: of the positives, the share flagged.

A32. Test error $= \text{bias}^2 + \text{variance} + \text{noise}$. With few rows and many correlated predictors, OLS is unbiased but its variance is enormous; a little bias that cuts the variance a lot lowers the total.

A33. It makes one in-or-out decision at a time, each depending on the previous ones; resampling the rows changes the path and the final model, and noise variables get in. The selection reflects this sample, not a property of the world, and the p-values computed afterwards are invalid.

A34. It shrinks every coefficient toward zero, more where predictors are collinear, but never to exactly zero. $X^\top X + \lambda I$ is invertible for any $\lambda > 0$, even when $X^\top X$ is singular ($p > n$).

A35. It sets weak coefficients to **exactly zero** (the corners of its diamond-shaped budget), so it selects variables. Limits: with $p > n$ it keeps at most n variables; among correlated predictors it keeps one, almost arbitrarily; the selected set is not "the true model".

A36. A ridge component: correlated predictors are kept or dropped **together** instead of one standing in for the group — the **grouping effect**.

A37. The penalty compares coefficient sizes, which depend on units: unstandardised, a variable in euros and one in thousands of euros are punished differently. Inside each fold, because standardising on all rows lets the held-out fold's information into the training, which makes the CV score optimistic (leakage).

A38. λ_{min} minimises the CV error. λ_{1se} is the largest λ whose error is within one standard error of that minimum. It gives a simpler (more shrunk) model whose error is indistinguishable from the best.

A39. FE remove everything constant within a country over time (geography, history, institutions); the cost is that the effect of any time-invariant variable cannot be estimated, and only within-country variation identifies the slopes. RE assume the country effects are uncorrelated with the regressors. The Hausman test compares the two; choose on whether the RE assumption holds, never on which gives the significant answer.

A40. Observations of the same country are correlated over time, so the sample carries less independent information than n suggests. Cluster the standard errors by country.

Part B — Calculations

B1. **(a)** Mean $= 35/5 = 7$; sorted $4, 6, 7, 8, 10$, median $= 7$. **(b)** Deviations $-3, 1, -1, 3, 0$; squares $9, 1, 1, 9, 0$, sum 20; $s^2 = 20/4 = 5$, $s = 2.236$. **(c)** $SE = 2.236/\sqrt{5} = 1.00$.

B2. **(a)** $SE = 12/\sqrt{36} = 2$; interval $50 \pm 1.96 \times 2 = (46.08,\ 53.92)$. **(b)** $n = 144$: $SE = 12/12 = 1$, halved; s stays about 12 — more data does not make the observations less spread out.

B3. **(a)** $g_1 = \frac{1}{36} \cdot \frac{1{,}166{,}400}{30^3} = \frac{1{,}166{,}400}{972{,}000} = 1.20$; threshold $2\sqrt{6/36} = 0.816 \rightarrow$ substantially **right**-skewed, consistent with mean > median. **(b)** $g_2 = \frac{1}{36} \cdot \frac{116{,}640{,}000}{30^4} - 3 = 4.00 - 3 = 1.00$; threshold $4\sqrt{6/36} = 1.633 \rightarrow$ not substantially kurtic. **(c)** The median (105): the right tail pulls the mean up.

B4. **(a)** $\bar{x} = 3$, $\bar{y} = 4$; $\sum(x-\bar{x})(y-\bar{y}) = 4+0+0+0+2 = 6$, $\sum(x-\bar{x})^2 = 10$; $\hat{\beta}_1 = 0.6$, $\hat{\beta}_0 = 4 - 0.6 \times 3 = 2.2$. **(b)** Fitted $2.8, 3.4, 4.0, 4.6, 5.2$; residuals $-0.8, 0.6, 1.0, -0.6, -0.2$ (they sum to 0); $SSR = 0.64+0.36+1+0.36+0.04 = 2.4$. **(c)** $SST = 4+0+1+0+1 = 6$; $R^2 = 1 - 2.4/6 = 0.60$; $RSE = \sqrt{2.4/3} = 0.894$.

B5. **(a)** $t = 0.85/0.34 = 2.50$; interval $0.85 \pm 0.666 = (0.184,\ 1.516)$; significant at 5% ($|t| > 1.96$; the interval excludes 0). **(b)** $\bar{R}^2 = 1 - 0.60 \times 49/45 = 0.347$.

B6. **(a)** Bias $= \beta_z \delta_1 = -0.3 \times 0.5 = -0.15$: negative, the short regression **understates** β_1. **(b)** $0.45 = \beta_1 - 0.15$, so $\beta_1 = 0.60$.

B7. **(a)** $\sum h_{ii} = p = 3$; average $3/40 = 0.075$. **(b)** Yes: 0.21 is 2.8 times the average, above the usual $2p/n = 0.15$ flag. High leverage alone is not influence: check its residual and its Cook's distance. **(c)** $4/n = 0.10$.

B8. $VIF = 1/(1-0.92) = 12.5$: the coefficient's variance is 12.5 times what it would be with uncorrelated regressors, its SE $\sqrt{12.5} \approx 3.5$ times larger. Above 10: a problem.

B9. **(a)** $AIC_A = 420+8 = 428$, $AIC_B = 410+14 = 424$; $BIC_A = 420+4\times4.605 = 438.4$, $BIC_B = 410+7\times4.605 = 442.2$. **(b)** AIC prefers B (lower), BIC prefers A. BIC charges $\ln 100 = 4.6$ per parameter instead of 2, so B's three extra parameters cost more than its gain in fit.

B10. **(a)** $x^\top\hat{\beta} = -3.0 + 0.04 \times 50 + 0.9 = -0.1$; odds $= e^{-0.1} = 0.905$; $p = 0.905/1.905 = 0.475$. **(b)** $e^{0.9} = 2.46$: a foreign-owned firm has 2.46 times the odds of exporting of an otherwise identical domestic firm (146% higher odds). **(c)** $p(1-p)\beta = 0.475 \times 0.525 \times 0.04 = 0.0100$: about **1.0 percentage point** per point of productivity, at this firm.

B11. **(a)** Odds $0.20/0.80 = 0.25$; new odds $0.25 \times 1.5 = 0.375$; new probability $0.375/1.375 = 0.273$. **(b)** +7.3 percentage points (from 20% to 27.3%), a 36% relative rise. An odds ratio multiplies the **odds**, not the probability.

B12. **(a)** $LR = 2(-412.6 + 418.9) = 12.6 > 7.81$ (χ^2_3): the three dummies jointly improve the model; reject the null that all three are zero. **(b)** $1 - 412.6/450 = 0.083$.

B13. **(a)** $\tau^\star = c_{FP}/(c_{FP}+c_{FN}) = 1/(1+9) = 0.10$. **(b)** $0.15 > 0.10$: flag. With 0.5 the applicant would not be flagged — the default threshold ignores the costs.

B14. $P(Y \le 1) = F(1.0 - 1.5) = F(-0.5) = 0.378$; $P(Y \le 2) = F(3.0 - 1.5) = F(1.5) = 0.818$. So $P(1) = 0.378$, $P(2) = 0.818 - 0.378 = 0.440$, $P(3) = 1 - 0.818 = 0.182$ (they sum to 1).

B15. Denominator $e^0 + e^{-0.5} + e^{0.2} = 1 + 0.607 + 1.221 = 2.828$. $P(\text{academic}) = 1/2.828 = 0.354$, $P(\text{general}) = 0.607/2.828 = 0.215$, $P(\text{vocation}) = 1.221/2.828 = 0.432$.

B16. **(a)** $TP = 30$, $FP = 70$, $FN = 20$, $TN = 880$. **(b)** Accuracy $= 910/1000 = 0.91$; precision $= 30/100 = 0.30$; recall $= 30/50 = 0.60$. **(c)** $950/1000 = 0.95$ — higher than the model's accuracy, although it finds no failure at all.

B17. **(a)** $\det = 24 - 9 = 15$; $(X^\top X)^{-1} = \frac{1}{15}\begin{pmatrix} 4 & -3 \\ -3 & 6 \end{pmatrix}$; $\hat{\beta}^{OLS} = \frac{1}{15}(48-30,\ -36+60) = (1.200,\ 1.600)$. **(b)** $X^\top X + 2I = \begin{pmatrix} 8 & 3 \\ 3 & 6 \end{pmatrix}$, $\det = 39$; $\hat{\beta}^{ridge} = \frac{1}{39}(72-30,\ -36+80) = (1.077,\ 1.128)$. **(c)** $1 - 1.077/1.2 = 10.3\%$ and $1 - 1.128/1.6 = 29.5\%$.

B18. **(a)** Subtract 0.25 from $|\rho_j|$, keep the sign, set to 0 if negative: $(0.35,\ -0.10,\ 0,\ 0,\ 0.03)$ — variables 1, 2 and 5 selected. **(b)** Threshold $\lambda\alpha = 0.125$, divisor $1 + 0.25 \times 0.5 = 1.125$: $(0.475,\ -0.225,\ 0.025,\ 0,\ 0.155)/1.125 = (0.422,\ -0.200,\ 0.022,\ 0,\ 0.138)$ — variables 1, 2, 3 and 5. **(c)** The elastic net keeps one more (variable 3): its lower threshold lets weaker predictors in, and its ridge part shrinks the survivors proportionally.

B19. $\lambda_{min} = 0.1$ (error 1.90). One-SE bound $1.90 + 0.11 = 2.01$. The largest λ with error ≤ 2.01 is 0.5 (1.98; at 1.0 the error 2.05 is above). $\lambda_{1se} = 0.5$: more shrinkage, so a smaller model, with an error indistinguishable from the best.

B20. **(a)** Effect $= 3.0 - 2 \times 0.05\,x = 3.0 - 0.1\,x$: +2.0 at 10 years, −1.0 at 40. **(b)** $x^\star = 3.0/0.1 = 30$ years (a maximum, since the square term is negative). **(c)** 3.0 is the effect only at $x = 0$; with a square term the effect depends

on x.

B21. **(a)** Men: 1.5 per year; women: $1.5 - 0.5 = 1.0$ per year. **(b)** Gap (women − men) $= 4 - 0.5\,\text{exp}$: +4 at 0 years, $4 - 5 = -1$ at 10. **(c)** $4 - 0.5\,\text{exp} = 0$ at 8 years.

B22. Classical: $t = 3.75$, interval $(0.143,\ 0.457)$. Clustered: $t = 1.67$, interval $(-0.053,\ 0.653)$. Report the clustered one: the panel's observations are not independent. The estimate did not change; what you may claim did — no longer significant at 5%.

B23. $14.2 > 7.81$ (χ^2_3): reject. FE and RE differ by more than sampling noise, so the RE assumption fails — use fixed effects.

B24. Four dummies; the omitted level is the reference. With five plus an intercept, the dummies sum to the intercept column: perfect collinearity (the dummy-variable trap), and the model cannot be estimated.

Part C — Diagnosis

C1. The empirical rule is applied to a variable that is clearly skewed: income cannot be negative, and an SD of 71% of the mean signals a long right tail. Report the median and the IQR instead, and check g_1 against its threshold.

C2. "Not significant" is read as "no effect". With $p = 0.31$ the data cannot distinguish the effect from zero, but the interval almost certainly also includes large negative values. Report the interval and say the evidence is inconclusive.

C3. Unbiased coefficients do not save the inference: under heteroskedasticity the classical standard errors, t-statistics and intervals are wrong. Use robust (HC) standard errors.

C4. AIC compared on different rows: 1,200 against 900. Likelihoods on different samples are not comparable, and the drop mostly reflects fewer observations. Refit both on the same 900 rows.

C5. An odds ratio read as a probability ratio (relative risk), and the conclusion is that collateral raises default risk. 1.40 means the **odds** are 40% higher; the change in probability depends on the baseline. (And a positive association with collateral may reflect which borrowers are asked to provide it — not causation.)

C6. Accuracy on a 1% outcome: "never fraud" is 99% accurate. Report the base rate, precision and recall. And the threshold ignores the costs: $\tau^\star = 1/(1+50) \approx 0.02$, not 0.5.

C7. Three errors. Standardising on **all** rows before cross-validation (leakage). Treating the lasso's selection as the true model. Reporting OLS p-values on the selected columns: the same data chose and then tested them, so the p-values are invalid. Fix: put the scaler inside a pipeline, report the selected set and its out-of-sample error, no post-selection p-values.

C8. The specification is chosen by which gives the significant result — specification search. Choose on the assumption: run the Hausman test (or argue whether the country effects can be correlated with the regressor). If RE is rejected, FE is the estimate, insignificant or not.

Part D — Judgement (what a strong answer contains)

D1. What it is: a cross-country association, €410 per index point, holding nothing else fixed that we know of. **Not licensed:** "will raise": richer countries can afford digital government (reverse causality), and institutions or education can drive both (omitted variables); $R^2 = 0.58$ says nothing about cause. **Evidence needed:** a design that isolates the change — within-country change over time with fixed effects, a staggered rollout compared with non-adopters, controls for the obvious confounders. **Meanwhile:** report the association with its interval, and do not attach a euro figure to the policy.

D2. What it is: a predictive association in this bank's data. **Not licensed:** region may stand in for income, sector or collateral; refusing a whole region also raises fairness and legal issues. **Evidence:** does the odds ratio survive controls for the borrower's own risk factors? Is the model calibrated for that region? **Meanwhile:** price risk on the borrower's characteristics, use a cost-based threshold, and review the variable's legitimacy before using it.

D3. What it is: six variables that jointly predict well at one λ, on this sample. **Not licensed:** a dropped variable is not "no effect" — the lasso keeps one of a correlated group almost arbitrarily, and with $p > n$ it can keep at most n. **Evidence:** selection stability across resamples, an elastic net to see whether groups enter together, a causal design for any variable whose effect matters. **Meanwhile:** report the predictive performance, not a list of "drivers".

D4. Fixed effects use only **within-firm** variation: the estimate answers "when **this** firm raises its training budget, does **its** productivity rise?" It removes everything constant about each firm, including whatever makes some firms both train more and be more productive. The question "are high-training firms more productive than others?" is a **between-firm** comparison that FE deliberately throws away; answering it needs the cross-sectional variation, with

its own confounding problems.

D5. The mean is pulled up by a few very large firms: $g_1 = 6.4$ is far above any threshold, and the median (18) is a fourteenth of the mean. The typical firm has about 18 employees — small, not medium. Report the median and the IQR, say the distribution is strongly right-skewed, and model the log of employment if it enters a regression.

Questions: `PRACTICE.md` · *the exam:* `EXAM.md` *and its solutions* · *the summary:* `MIDTERM-SUMMARY.pdf`